

Training Day – 13

Today's training focused on **CI/CD, software deployment strategies, and DevOps practices**. The session provided an understanding of how modern software development teams automate the process of building, testing, and deploying applications while maintaining consistency across different environments.

Work Undertaken:

- Learned the fundamentals of **CI/CD (Continuous Integration and Continuous Delivery/Deployment)** and understood how it helps automate and improve software development and deployment workflows.
- Learned about the challenges associated with **manual software deployment**, including human errors, inconsistent configurations, repetitive tasks, and increased deployment time.
- Learned the fundamentals of **DevOps**, including its lifecycle and its role in improving collaboration between development and operations teams.
- Compared different **deployment strategies**, including:
 - **Blue-Green Deployment:** Maintains two environments and switches traffic between them to reduce downtime.
 - **Canary Deployment:** Gradually releases a new version to a small group of users before expanding the release.
 - **Rolling Deployment:** Replaces application instances gradually instead of updating all instances simultaneously.

Additional Learning:

The session helped me understand how CI/CD and DevOps practices improve the software development lifecycle by reducing manual effort, increasing deployment reliability, and enabling faster delivery of software. Comparing different environments and deployment strategies also provided insight into how organizations manage application releases with minimal disruption.

Conclusion:

The day provided a strong foundation in **CI/CD and DevOps concepts**. Understanding pipeline stages, deployment environments, release strategies, and the CAMS principles improved my knowledge of modern software development and deployment practices.